Assign2_1.java

Oop.java > Assignments.java > src > Assign2_1.java > Assign2_1 > main(String[])

```java
// Create a Java program using bitwise operators that will produce the output: true, 10,
// false, 15.

public class Assign2_1{
    public static void main (String[] args)
    {
        // BITWISE
        int a = 7; // 0111
        int b = 1; // 0001
        int result_ten = (b << 3) | (b << 1); //8 | 2
        int result_fifteen = (a << 1) | b; // 1110 | 0001 = 15 (1111)
        boolean true_result = (result_ten & 1) == 0;
        boolean false_result = (result_ten & 1) == 1;

        // PRINT
        System.out.println((true_result));
        System.out.println((result_ten));
        System.out.println((false_result));
        System.out.println((result_fifteen));
    }
}
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

Code - src

```
cd "/home/rai/Documents/Programming/Back End/Java/Oop.java/Assignments.java/src/" && javac Assign2_1.java && java Assign2_1
pop-os% cd "/home/rai/Documents/Programming/Back End/Java/Oop.java/Assignments.java/src/" && javac Assign2_1.java && java Assign2_1
true
10
false
15
pop-os%
```

Assign2_2.java

Oop.java > Assignments.java > src > Assign2_2.java > Assign2_2 > main(String[])

```java
//Create a Java program using the ternary operator that will produce the output: true,
// false, 25, 30.

public class Assign2_2 {
    public static void main (String[] args)
    {
        // TERNARY OPERATORS
        int a = 20;
        int b = 5;
        int c = 10;

        // TRUE - FALSE
        boolean result_true = (a > b) ? true : false;
        boolean result_false = (a < b) ? true : false;
        System.out.println(result_true);
        System.out.println(result_false);

        // 25 - 30
        System.out.println ((a == 25) ? 25 : a + b);
        System.out.println ((a == 30) ? 30 : a + c);
    }
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

Code - src

```
cd "/home/rai/Documents/Programming/Back End/Java/Oop.java/Assignments.java/src/" && javac Assign2_2.java && java Assign2_2
pop-os% cd "/home/rai/Documents/Programming/Back End/Java/Oop.java/Assignments.java/src/" && javac Assign2_2.java && java Assign2_2
true
false
25
30
pop-os%
```

Assign2_3.java

Oop.java > Assignments.java > src > Assign2_3.java > Assign2_3 > main(String[])

```java
// Create a Java program using assignment operators that will produce the output:
// Addition: 120, Subtraction: 70, Multiplication: 144, Division: 16.

public class Assign2_3 {
    Run | Debug
    public static void main (String[] args)
    {
        // ASSIGN
        int a = 100;
        int b = 2;
        int addition = 20;
        int subtraction = 170;
        int multiplication = 12;
        int division = 32;

        addition += a;
        subtraction -= a;
        multiplication *= multiplication;
        division /= b;
        // PRINT
        System.out.println(addition);
        System.out.println(subtraction);
        System.out.println(multiplication);
        System.out.println(division);
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS Code - src

```
cd "/home/rai/Documents/Programming/Back End/Java/Oop.java/Assignments.java/src/" && javac Assign2_3.java && java Assign2_3
pop-os% cd "/home/rai/Documents/Programming/Back End/Java/Oop.java/Assignments.java/src/" && javac Assign2_3.java && java Assign2_3
120
70
144
16
pop-os%
```

Assign2_3.java | Assign2_4.java

Oop.java > Assignments.java > src > Assign2_4.java > Assign2_4 > main(String[])

```java
public class Assign2_4 {
    Run | Debug
    public static void main (String[] args)
    {
        int var1 = 5;
        int var2 = 7;
        int var3 = 9;
        int temperature = 30;
        int average_temp = 25;
        boolean is_above_average;
        int score = 40;
        boolean passed_score = score >= 50;

        //Operators
        int arithmetic = var2 % var1;
        int even = var3;
        // NUMERIC
        System.out.println(arithmetic); // Arithmetic for positive num
        System.out.println(var1++ - --var3); // Tricky Increment for negative num

        if (var1 > var2){ // Relational for finding higher values
            System.out.println(var1);
        } else{
            System.out.println(var2);
        }

        while (true) { // Relational, Arithmetic, and Assignment for printing even num
            if (even % 2 == 0){
                System.out.println(even);
                break;
            }
            else {
                even += 1;
```

Assign2_3.java   Assign2_4.java

Oop.java > Assignments.java > src > Assign2_4.java > Assign2_4 > main(String[])

```java
public class Assign2_4 {
    public static void main (String[] args)
                    break;
                }
                else {
                    even += 1;
                }
            }

            // BOOLEANS
            if (temperature > average_temp){ // Boolean to check if temp is above average
                is_above_average = true;
            } else {
                is_above_average = false;
            }
            System.out.println(is_above_average);

            System.out.println(passed_score); // Boolean to check passing score
        }
}
```

PROBLEMS   OUTPUT   DEBUG CONSOLE   TERMINAL   PORTS   Code - src

```
cd "/home/rai/Documents/Programming/Back End/Java/Oop.java/Assignments.java/src/" && javac Assign2_4.java && java Assign2_4
pop-os% cd "/home/rai/Documents/Programming/Back End/Java/Oop.java/Assignments.java/src/" && javac Assign2_4.java && java Assign2_4
2
-3
7
10
true
false
pop-os%
```